



Investigating the Impact of Meteorological Parameters on Surface Ozone Variability in Nigeria: A Principal Component and Regression Analysis Approach

Ademola J. Oyewole^{1*}, Emmanuel F. Nymphas²

¹ College of Natural Sciences, Physics Unit, Westland University, 232101 Iwo, Nigeria

² Department of Physics, University of Ibadan, 200273 Ibadan, Nigeria

* Correspondence: Ademola J. Oyewole (joeademola@gmail.com)

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Abstract: Surface ozone is a major secondary air pollutant carrying significant implications for the stability of the ecosystem, public health, and climate forcing. In Nigeria, rapid urbanization, emission growth, and climatic variability interact to shape ozone dynamics; however, there is little understanding of the specific contribution of meteorological factors. This study investigated the influence of key meteorological variables on surface ozone variability in Nigeria based on a combined Principal Component Analysis (PCA) and Linear Regression (LR) framework. Meteorological parameters include temperature, wind speed, relative humidity, rainfall, cloud cover, short-wave radiation, and atmospheric pressure. PCA results revealed that the first three principal components (PCs) explained approximately 76% of the total variance, suggesting that a limited number of climatic modes, primarily associated with temperature, wind speed, and humidity, dominated ozone variability. LR further quantified the individual contributions of these variables. Cloud cover (-1.015), temperature (-0.975), and wind speed (-0.665) exerted the strongest negative influence on surface ozone concentrations. In contrast, rainfall (0.306) demonstrated a positive association, potentially linked to enhanced post-precipitation soil NO_x emissions. Other variables, including short-wave radiation, atmospheric pressure, and relative humidity, exhibited relatively minor effects. While the model reflected robust predictive performance (Mean Squared Error (MSE) = 0.044), the findings emphasized the significant role of meteorological processes in shaping ozone variability. In drastically urbanizing tropical regions, meteorological dynamics should be incorporated into the forecast of air quality and planning of environmental policies.

Keywords: Surface ozone; Meteorological drivers; Principal component analysis; Linear regression; Nigeria

1 Introduction

The impacts of tropospheric ozone, or atmospheric ozone near the earth's surface, on respiratory ailments, vegetation damage, and climate forcing have become a growing public health and environmental issue. Rapid urbanization, rising vehicle emissions, and industrial growth all contribute to Nigeria's ozone concentrations, although there is insufficient reference on how exactly these meteorological factors are measured. Designing efficient regulatory regulations and predicting air quality under fluctuating climatic circumstances help examine how factors like temperature, humidity, short-wave radiation, wind speed, rainfall, and cloud cover influence ozone variability.

Several recent studies globally have emphasized the importance of meteorology in modulating ozone levels. For example, Li et al. [1] analyzed surface ozone trends across China and assessed the uncertainties introduced by using different meteorological datasets and analytical methods. They found that meteorological conditions significantly amplify or dampen observed ozone trends. Likewise, multi-sensor reanalysis data were employed to assess long-term trends in total column ozone over Nigeria, thus revealing spatial and interannual variability associated with climatic parameters [2]. These studies underscored both the utility and necessity of combining observational data with statistical techniques to disentangle complex drivers.

In Nigeria, a few studies have begun to address ozone and its health implications, but some research gaps remain to be bridged. For instance, Ihedike et al. [2] conducted real-time monitoring of ground-level ozone in Abuja, revealing substantial diurnal and seasonal variability, especially during Harmattan, but did not fully isolate the meteorological controls using multivariate statistical models. Meanwhile, works focusing on urban air quality and air pollutant concentrations often included meteorological parameters but seldom applied dimensionality reduction tools like Principal Component Analysis (PCA), in combination with regression to robustly identify dominant drivers. A methodological framework is indispensable as it could not only quantify relationships, but also reduce redundancy among predictors and capture underlying structure.

This study aims to address these gaps by integrating PCA with Linear Regression (LR) to analyse the relationship between meteorological variables and surface ozone concentrations in Nigeria. The PCA allows extraction of dominant modes or patterns among meteorological variables, thus reducing multicollinearity and helping to understand the major dimensions that influence ozone variability. Then, regression will quantify the contribution of each meteorological factor (or principal component) to changes in the observed ozone concentration. Such a dual approach enhances interpretability and statistical reliability, especially in complex environments where many interrelated meteorological variables might obscure individual effects.

Lastly, given Nigeria's susceptibility to climate change and the impact of air pollution on human health, the results of this investigation had their relevancy and applications in both science and practice. Scientifically, they helped build knowledge about how climate and weather interacted with air chemistry in tropical regions; practically, policy makers and environmental agencies could obtain insights for forecasting, setting emission standards, and planning air quality interventions (e.g., warning systems during high ozone episodes).

2 Methodology

2.1 Data Collection

In order to capture variability, the study employed Satellite Monthly ground-level ozone concentration data for Nigeria from the National Aeronautics and Space Administration (NASA) website (<https://giovanni.gsfc.nasa.gov>) for the years 2002–2024, i.e., 23 years. During this period, ozone and the following meteorological variables were included in the data: temperature, relative humidity, cloud cover, rainfall, wind speed, air pressure, and short-wave radiation. The data included seasonal cycles. Data preparation took into account of three considerations: Temporal alignment (to ensure meteorological and ozone data have the same time steps, such as hourly or monthly), quality control (to remove missing or blatantly incorrect values), and normalization or standardization of continuous variables to enable comparison.

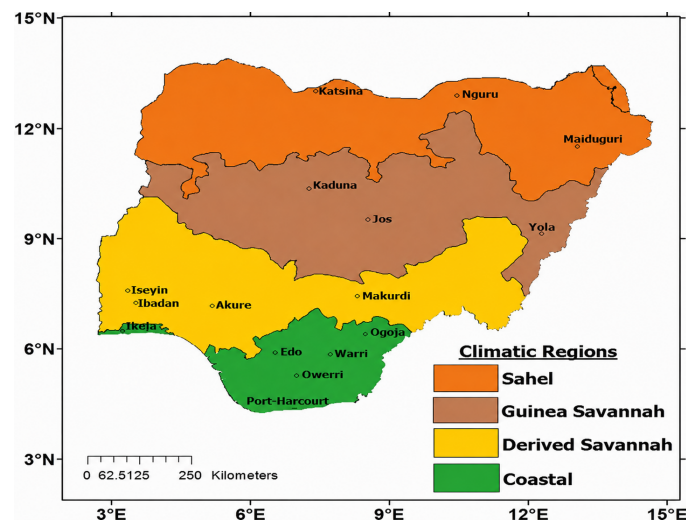


Figure 1. Map of Nigeria with major climatic regions and few major cities

With a population of more than 230 million, Nigeria is the most populous nation in Africa and has the biggest economy on the continent [3, 4]. The nation, which has a land area of around 923,768 km², is located in West Africa between latitudes 4°N and 14°N and longitudes 3°E and 15°E, as shown in Figure 1. The Atlantic Ocean (Gulf of Guinea) constitutes Nigeria's southern border, while Niger, Chad (across Lake Chad), Cameroon, and the Benin Republic border it on the north, east, and west, respectively [5]. The advantageous geographic location of Nigeria placed it inside a tropical climate belt that is heavily impacted by both continental and marine air masses.

The climate in Nigeria reflects notable north-south gradients. While the northern Sahelian edge receives less than 500 mm of annual rainfall, due to periodic droughts and the effect of dry and dusty Harmattan winds, the

southern coastline region enjoys a wet tropical environment with rainfall surpassing 3,000 mm yearly. Variations in the Intertropical Convergence Zone (ITCZ), which regulates temperature and precipitation patterns nationwide, are indicative of seasonal transitions [6]. Temperature, humidity, precipitation, and radiation gradients of this kind offer a natural laboratory for studying how weather affects ozone variability.

Concerns over air quality have intensified due to Nigeria's fast urbanization, with over 54% of the population in the country already settling in cities [4]. Vehicle traffic, industry, power generation, and biomass burning all contribute to the high levels of ozone precursor emissions in cities like Lagos, Abuja, Port Harcourt, and Kano. Urban Nigeria is especially susceptible to ozone-related health hazards because of the interaction between these human-caused factors and regional climatic circumstances [7].

This challenge was addressed by previous studies, which acknowledged that the maximum daily 8-hour average ozone threshold of 70 parts per billion by volume (ppbv) was frequently exceeded in Nigerian locations [8]. They attributed the amount to a combination of burning biomass, industrial processes, and fossil fuels. Seasonal and diurnal ozone maxima linked to climatic variability were revealed by more focused studies conducted in Abuja and Kano [2]. In order to determine the most significantly meteorological causes of ozone fluctuation throughout Nigeria's various geographical and climatic zones, these data were subjected to a thorough statistical investigation with techniques like PCA and regression.

2.2 Data Processing

NASA provided surface ozone concentration data and related meteorological factors for the Nigerian site. Among the meteorological factors taken into account, the study involved air temperature, relative humidity, short-wave solar radiation, wind speed, rainfall, atmospheric pressure, and cloud cover. These factors are all known to have an impact on the processes of ozone generation, transport, and removal. Satellite-based observations have been widely adopted to assess anthropogenic emissions and their implications for ozone pollution in Nigeria [8]. To guarantee temporal continuity and spatial consistency, data were taken from quality-controlled observational and/or reanalysis datasets. Screening for missing values, outliers, and physically implausible findings was part of the initial data processing. In accordance with the accepted atmospheric data-handling protocols, missing data points were handled using standard gap-filling techniques, such as linear interpolation for short gaps or elimination, when data gaps exceeded acceptable criteria [9].

Each meteorological variable was standardized by z-score normalization, which is defined as follows:

$$Z = (X - \mu) / \sigma \quad (1)$$

where, X is the original variable, μ is its mean, and σ is its standard deviation, to prevent scale bias in multivariate analysis. Larger numerical range variables were prevented from controlling the principal component structure via standardization [10].

2.3 Variable Selection

Both atmospheric chemistry theory and earlier empirical research on ozone variability served as a reference for the selection of meteorological variables. Wind speed affects horizontal dispersion and ventilation, whereas temperature and sunlight increase the rates of photochemical reactions that produce ozone. While cloud cover modifies incoming solar energy, relative humidity and rainfall have an impact on ozone through cloud formation, moist deposition, and suppression of photochemical activity. Synoptic-scale circulation patterns, which might affect stagnation or pollutant movement, are reflected in atmospheric pressure [11–14]. To evaluate the interdependencies between predictors and their connections to surface ozone concentration, exploratory correlation analysis was carried out prior to PCA. The employment of PCA as a dimensionality-reduction method was justified by strong correlations between meteorological variables, which verified the existence of multicollinearity. To improve the interpretability and robustness of the model, the study eliminated variables with insignificant variance or poor physical significance to ozone chemistry.

2.4 Principal Component Analysis

The standardized meteorological dataset was subjected to PCA in order to identify the primary modes of variability controlling the weather environment that affects surface ozone. The original set of correlated variables is converted by PCA into a new set of orthogonal (uncorrelated) principal components (PCs), each of which is a linear combination of the original variables [11].

The PCA procedure undertook the following steps:

1. Construction of the covariance (or correlation) matrix from standardized meteorological variables;
2. Eigenvalue decomposition of the matrix to obtain eigenvalues and eigenvectors;
3. Selection of significant components based on the Kaiser criterion (eigenvalues > 1) and visual inspection of the scree plot;

4. Interpretation of component loadings, where highly positive or negative loadings indicate strong contributions of specific meteorological variables to each principal component;
5. Rotation, if applied, use varimax rotation to enhance physical interpretability of the components;
6. Independent climatic regimes (such as photochemically favorable circumstances, wet-season suppression, or ventilation-dominated regimes) are represented by the preserved primary components. For the purposes of reducing multicollinearity and enhancing statistical stability, these PCs were later employed as predictors in regression modeling [15].

2.5 Regression Modelling

LR modeling was used to measure the impact of weather on surface ozone variability. Two methods of regression were taken into consideration:

1. Principal Component Regression (PCR), which uses specific PCs as predictors, and
2. Direct Multiple Linear Regression (MLR), which applies standardized meteorological variables.

$$O_3 = \beta_0 + \sum_{i=1}^n \beta_i x_i + \varepsilon \quad (2)$$

where, β_i are regression coefficients, and ε is the residual error term. The coefficient of determination R^2 , adjusted R^2 , root mean square error (RMSE), and statistical significance (p -values) of regression coefficients were used to assess the effectiveness of the regression model. To validate important regression assumptions such as residual normality, homoscedasticity, and the lack of autocorrelation, diagnostic tests were carried out.

Regression analysis omitted multicollinearity by employing PCA-derived predictors, which rendered it easier to attribute ozone fluctuation to distinct meteorological sources. This PCA-regression framework has been used extensively in studies on climate and air quality, and it has been successful in separating the intricate relationships between atmospheric contaminants and meteorology [10, 15].

2.6 Extraction of Key Meteorological Drivers Using Principal Component Analysis

To minimize dimensionality and identify PCs that account for the bulk of variation in the meteorological dataset, PCA would be performed to the collection of meteorological variables following preprocessing. Besides, to determine which initial climatic factors contributed most to each component, the PCs would then be interpreted by their respective loadings. For instance, a PC could load significantly on temperature, solar radiation, and cloud cover. This helped reduce multicollinearity in regression and advanced insight into underlying meteorological modes that might be influencing ozone.

2.7 Linear Regression and Model Assessment

Following the derivation of PCs, regression analysis would be performed in two ways: first, ozone would be regressed on the PCs to investigate the combined influence of the major modes; second, observed ozone concentrations would be regressed directly on the original meteorological variables to estimate their individual effects. Once the model assumptions (linearity, uniformity, independence, multicollinearity, and residual normality) were confirmed, the regression model(s) might be MLR (ordinary least squares). Methods like stepwise regression or information criteria might be employed for model selection.

Metrics like Mean Squared Error (MSE) would be used to evaluate the performance of the model. Contextualizing the data would be aided by comparison with previous research, if available, conducted in Nigeria or other tropical environments.

3 Discussion of Principal Component Analysis and Linear Regression Results

3.1 Principal Component Analysis Results

According to the results of PCA, the top three PCs accounted for over 76% of the variation (30.1%, 24.9%, and 21.1%, respectively). The results are explicitly listed in Table 1 and Figure 2.

Table 1. Variance of each meteorological principal component (PC)

PC	Variance
Temperature	0.301
Wind speed	0.249
Humidity	0.211
Pressure	0.092
Rainfall	0.077
Short-wave radiation	0.044
Cloud cover	0.025

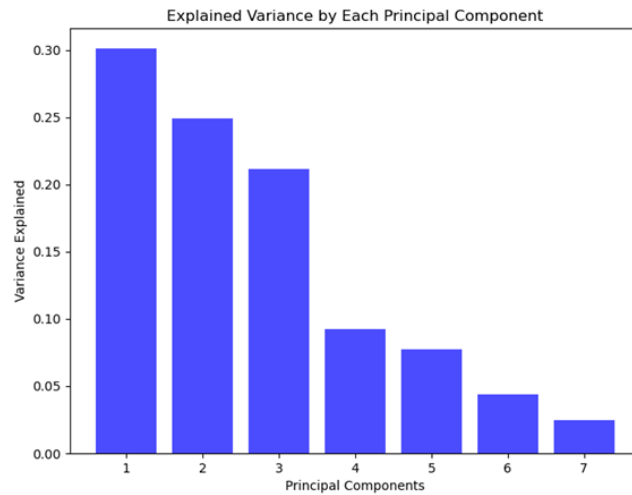


Figure 2. Variance of ozone as explained by each component

Note: 1-temperature, 2-wind speed, 3-humidity, 4-pressure, 5-rainfall, 6-short wave radiation, 7-cloud cover

Instead of having all parameters acting equally, a small number of major meteorological variables collaborated to produce surface ozone variability in Nigeria. The comparatively sharp decline in explained variance following the third component implied that the first few dimensions accounted for the majority of significant variability. This result is consistent with earlier atmospheric research that used PCA to separate the impact of intricate meteorological factors on air quality. In tropical and subtropical climates, for example, temperature, humidity, and radiation tended to cluster strongly as the main variables influencing ozone generation [15]. In a similar vein, Marais et al. [8] pointed out that, in West African settings, despite a variety of contributing variables, a small number of meteorological modes might adequately explain ozone fluctuation.

Thus, the PCA strengthened the case that Nigerian surface ozone variability was governed by a few critical meteorological interactions, consistent with the research focus on identifying dominant drivers.

3.2 Linear Regression Results

LR results underpinned that meteorological parameter exerted varying degrees of positive and negative influence on surface ozone concentration.

Negative contributors: as shown in Table 2, the most potent negative drivers were wind speed (-0.665), temperature (-0.975), and cloud cover (-1.015). This implies that while higher wind speed promotes dispersion and lowers local concentrations, high cloud cover and reduced sun energy inhibit photochemical ozone production. According to the study in [16], the negative correlation between temperature and ozone build-up might be a reflection of the local environment, where high humidity and convective processes during hot seasons inhibited ozone build-up.

Positive contributor: also evident from Table 2, rainfall (0.306) exhibited a positive influence on surface ozone. In line with [17], this might be attributed to post-precipitation increases in ozone precursors, particularly soil NO_x emissions.

Table 2. Linear regression (LR) coefficients (meteorological parameters contribution)

Parameters	Coefficient
Temperature	-0.975
Wind speed	-0.665
Relative humidity	-0.190
Pressure	-0.143
Rainfall	0.306
Short-wave radiation	-0.063
Cloud cover	-1.015

Note: Mean Squared Error (MSE): 0.044

Other parameters: short-wave radiation (-0.063) was somewhat negative, whereas humidity (-0.190) and pressure (-0.143) displayed lesser but nonetheless negative correlations. Although this deviates from worldwide trends, it

could draw attention to the local atmospheric chemistry in Nigerian cities that is impacted by significant aerosol loading [18]. As seen in Figure 3 below, the comparatively low MSE (0.044) reflected by the model suggested a very strong fit, hence confirming the accuracy of these coefficients in capturing local ozone–meteorology interactions.

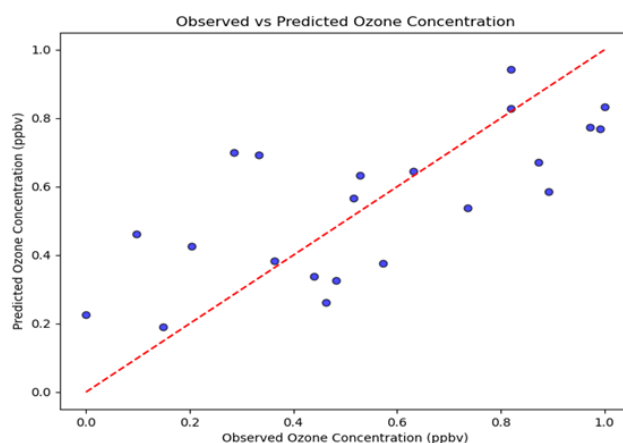


Figure 3. Scattered plots representing the observed and predicted ozone concentration

While the LR model specified the influence derived from the direction and size of each factor, the PCA findings successfully decreased dimensionality, thus indicating the key meteorological combinations driving ozone. When combined, both techniques offered complimentary insights: LR confirmed the precise function of each parameter, while PCA revealed the underlying variance structure. This dual strategy is in line with accepted atmospheric science procedures [16, 19], in which dominating patterns were identified using PCA and their effects were quantified using regression. The results confirmed the scientific and policy significance of this study by demonstrating that surface ozone fluctuation in Nigeria was not random; rather, it was regularly correlated with important meteorological causes.

4 Conclusions

PCA and LR were combined to provide powerful insights into the dynamics of surface ozone changes in Nigeria. The PCA results enriched understanding of a small number of climatic modes which accounted for the majority of variability in the meteorological dataset. The key roles played by temperature, humidity, and radiation interactions were emphasized in the determination of ozone patterns as temperature, wind speed, and cloud cover could all significantly reduce surface ozone concentrations. On the other hand, rainfall could only slightly increase the concentrations in accordance with the LR study. When the impacts of climatic elements were assessed individually, local air conditions and aerosol loading in Nigerian urban settings may be reflected in the comparatively constrained amount of short-wave radiation.

What the results imply is, instead of random occurrences, surface ozone variability in Nigeria is closely associated with quantifiable meteorological processes. These findings lend some support to the value of integrated multivariate statistical approaches for enhancing air-quality evaluation, forecasting, and environmental management in markedly urbanizing tropical regions and offer an advanced examination of the climatic influences on ozone evolution.

Author Contributions

Conceptualization, A.J.O. and E.F.N.; investigation, A.J.O.; writing-original draft preparation, A.J.O.; writing-review and editing, E.F.N.; supervision, E.F.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data [NetCDF] supporting our research results were deposited in NASA website. The data can be accessed at <https://giovanni.gsfc.nasa.gov>.

Conflicts of Interest

The authors declare no conflicts of interest.

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